

Panagiotis Aigaias Dimtsoudis

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Education

University of Pennsylvania – B.S.E. Electrical Engineering

Class of 2028

Relevant Coursework: Electrical Circuits and Systems, Electronic, Photonic, and Electromechanical Devices, Signal and Information Processing, Principles of Optics and Photonics, Embedded Systems Laboratory, Fundamentals of Solid State Circuits

SAT: 1500/1600 (Math: 780, Reading & Writing: 720)

Project Portfolio

Metal Detector PCB

- Designed and laid out an analog metal detector PCB using ALD1103 MOSFETs, LC oscillators, a mixer, filters, and amplifier stages.
- Verified oscillator frequencies, bias points, gain, and output behavior through hand calculations and LTspice simulations.
- Assembled, soldered, and tested the PCB, using oscilloscope measurements to debug each stage and drive an 8Ω speaker.

Wireless ECG Monitor (Analog Front-End + BLE)

- Built a portable ECG front-end using an instrumentation amp, virtual ground biasing, and band-limited filtering (~ 0.5 – 160 Hz)
- Sampled ECG at ~ 200 Hz on a Feather MCU and streamed it over BLE for live plotting on a central device
- Wrote Python scripts for logging/plotting and R-peak detection to estimate heart rate (BPM) from ECG data

Guitar Hero Synthesizer

- Repurposed a Guitar Hero controller into a playable embedded synthesizer using an ATmega328PB
- Wrote bare-metal C firmware for real-time fret, strum, mute, joystick, whammy, and LCD interface control
- Implemented PWM audio synthesis with power-chord playback, vibrato, and decaying note behavior
- Debugged power and audio issues by redesigning the speaker/power regulation path for reliable operation

Mesh Network for Hiking Trails

- Optimized Arduino/XBee mesh placement for ~ 100 – 150 m hops and multi-mile coverage
- Implemented info/distress modes with routing and ACKs for reliable delivery
- Made low-cost prototype/test plan for range, packet success rate, and power draw

Experience

Device Research and Engineering Laboratory

Philadelphia, PA

Research Assistant

May 2026 – Present

- Conduct cryogenic testing and electrical characterization of AlScN ferroelectric diode (Fe-Diode) devices to study low-temperature behavior, switching performance, and device reliability.
- Support PhD-led research on AlScN-based electronic and memory devices through literature review, cleanroom workflow preparation, and nanofabrication-related experimental work.
- Analyze measurement data from experimental devices to organize results, identify performance trends, and guide future characterization plans.

Technology Support Services

Philadelphia, PA

Technology Support Specialist

Jan 2025 – Present

- Provide software/hardware support to Penn undergraduates through in-person help and ticketing.
- Triage technical issues, document cases, coordinate escalations, and support equipment loaner operations.

Automation Systems Hellas

Thessaloniki, Greece

Computer Engineering Intern

Jun 2024 – Aug 2024

- Updated control panel documentation by labeling field wiring and tracking I/O device inventories.
- Supported OT and IT tasks by imaging PCs and HMIs and monitoring servers and network devices, including backup and access administration.
- Wrote C and Python utilities for log parsing and configuration checks to streamline commissioning preparation.

Academics & Activities

Science and Technology Wing (STWing)

Philadelphia, PA

Member

Sep 2024 – Present

- Built and debugged drone/blimp prototypes through subsystem wiring, bench testing, and flight validation.

Robotics Club / F1 in Schools

Thessaloniki, Greece

IT Lead

Sep 2021 – May 2024

- Led website/outreach and contributed to CAD design/testing for a nationally qualifying F1 team.

Skills

Programming & Embedded Systems: Python, Java, OCaml, C/C++, Assembly, UART/I2C, oscilloscopes, soldering

CAD/EDA: Altium Designer, LTspice, SolidWorks, Fusion 360

Interests: Embedded systems, PCB/chip layout, analog circuits, nanofabrication, ferroelectric memory

Languages: Greek (Native), English (Fluent), German (Intermediate), Japanese (Intermediate)